

# ASSIGNMENT : 5

**TITLE :** Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

**Problem Statement :** Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

**Learning Objectives :** To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

**Theory :** ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

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## Exercise :

1. Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

```
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {

        ArrayList<String> tasks = new ArrayList<>();

        tasks.add("Complete Java Assignment");
        tasks.add("Study Collections");
        tasks.add("Submit Lab Manual");

        StringBuffer sb = new StringBuffer();
        sb.append("To-Do List:\n");

        for (int i = 0; i < tasks.size(); i++) {
            sb.append((i + 1) + ". " + tasks.get(i) + "\n");
        }

        System.out.println(sb);
    }
}
```

Output

Generated files

```
To-Do List:
1. Complete Java Assignment
2. Study Collections
3. Submit Lab Manual
```

2. Create a **Student Course Registration System** using **ArrayList** to store the list of courses registered by a student and **StringBuffer** to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {

        ArrayList<String> courses = new ArrayList<>();

        courses.add("Programming with Java");
        courses.add("Data Structures");
        courses.add("Database Management");

        courses.remove("Data Structures");

        StringBuffer sb = new StringBuffer();
        sb.append("Registered Courses:\n");

        for (String course : courses) {
            sb.append(course).append("\n");
        }

        System.out.println(sb);
    }
}
```

Output

Generated files

```
Registered Courses:  
Programming with Java  
Database Management
```

## Outcome

- Understood the use of **ArrayList** for dynamic data storage.
- Implemented **StringBuffer** for efficient string manipulation.
- Learned to add, remove, and display dynamic data using Java Collections.

## Conclusion

This assignment demonstrated the use of **ArrayList** for storing and managing dynamic data and **StringBuffer** for efficient string manipulation. These classes simplify data management, improve performance, and are widely used in Java applications requiring dynamic collections and mutable strings.